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Research article

CentralBankRoBERTa: A fine-tuned large language model for central bank communications☆

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ABSTRACT

Central bank communications are an important tool for guiding the economy and fulfilling monetary policy goals. Natural language processing (NLP) algorithms have been used to analyze central bank communications. These outdated bag-of-words methods often ignore context and cannot distinguish who these sentiments are addressing. Recent research has introduced deep-learning-based NLP algorithms, also known as large language models (LLMs), which take context into account. This study applies LLMs to central bank communications and constructs CentralBankRoBERTa, a state-of-the-art economic agent classifier that distinguishes five basic macroeconomic agents and binary sentiment classifier that identifies the emotional content of sentences in central bank communications. The absence of large-language models in the central bank communications literature may be attributed to a lack of appropriately labeled datasets. To address this gap, we introduce our model, CentralBankRoBERTa, offering an easy-to-use and standardized tool for scholars of central bank communications.

1. Introduction

Today, central bank communications are considered an important monetary policy tool. Through communications, central banks have been said to guide the economy. Communications are supposed to help central banks better fulfill monetary policy goals such as price stability, stable interest rates or employment. A burgeoning literature has explored the use of natural language processing (NLP) to analyze central bank communications. Most, if not all, of these studies rely on NLP algorithms that assume a bag-of-words structure. That is, the algorithms ignore context and instead analyze texts as a collection of individual words treated independently without regard for grammar or word order. More recently, research in computational linguistics has introduced deep-learning-based NLP algorithms often referred to as large language models (LLMs) due to their large number of parameters (up to billions). These algorithms learn the syntactic and semantic relationships between words from a large body of texts. This allows them to take context into account.

The aim of this study is to apply LLMs to central bank communications. Using a sample of 15,060 manually prelabeled sentences from the Federal Reserve Bank (Fed), the Bank of International Settlements (BIS) and the European Central Bank (ECB), we test for the

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appropriate NLP method and construct the first economic agent classifier of central bank communications. This classifier distinguishes five basic macroeconomic agents in central bank communications: households, firms, the financial sector, governments, and the central banks themselves. This basic classification follows the institutional sectors of the System of National Accounts (2008) (SNA 2008).[1] It is our view that such a classifier is fundamental for analyzing central bank communications. Central bank communications do not take place in a vacuum. Monetary policy may address different, and sometimes even opposing agents within the economy, so that signals that are relevant to one agent may be adversely relevant to another, or not relevant at all. Research that is interested in better understanding how central bank communications address different actors within the economy; whether communications exhibit a bias of some actors over others; and which actors in the economy are paying the closest attention to what a central bank says may find such a classifier particularly useful.

Next, we may want to know whether a given communication emits a positive or negative signal about, and arguably to, the economic actor that a central bank is talking about. For this purpose, we test for the appropriate NLP method using a sample of 13,458 prelabeled sentences from the Fed, ECB and BIS, and construct a binary sentiment classifier. The classifier is able to distinguish sentences with regards to their emotional content. A word of caution on the interpretation of sentiment. In the literature, the emotional content of central bank communications is often equated with monetary policy stances. Positive emotions are then supposed to represent "doves", i.e. central bankers in favor of accommodative monetary policies, and negative emotions represent "hawks", i.e. central bankers in favor of restrictive monetary policies. While it may be generally useful to detect hawkish and dovish stances in central bank communications, we believe that it is misleading to attribute binary emotions to such stances. From a theoretical point of view, there is no reason to believe that a "hawkish" central banker will only talk "negatively" in all contexts. Indeed, if soft speech is like oil on bruised skin, hawkish central bankers would surely be well advised to speak positively on occasion without ever having to compromise their policy stance. Our sentiment classifier may be useful for research interested in monitoring how central banks feel about the constituents of the economy and whether these feelings manage to steer the economy in specific directions or not.

The rest of this paper is structured as follows. Section 2 reviews some of the recent literature of central bank communication with a focus on the most common practices of natural language processing. Section 3 gives an overview of our data and labeling method. Section 4 presents an experiment to test what NLP method is best suited for our two classification tasks. We compare different state-of-the-art LLMs such as BERT, FinBERT, RoBERTa and XLNet as well as more traditional machine learning algorithms such as LSTMs, Naive Bayes, Support Vector Machines or Random Forest. We conclude our paper with a few suggestions for the type of analysis our model could be used for.

2. Measuring central bank communication

Empirical analyses of central bank communication have relied on natural language processing (NLP) techniques to quantify textual information. Most studies employ NLP for downstream econometric purposes, which may be the reason why less attention has been given to studying the accuracy of the linguistic models employed. In our view, NLP models that have been used so far in measuring central bank communications have failed on two fronts. First, they are unable to take into consideration the contextual information of central bank communications and, second, they do not view communication as an interaction between a speaker and an audience. The first problem would be less severe if central bank communications were linguistically simple. Alas, they are not. The Flesch-Kincaid score, a test giving information on the easiness of understanding a text, of an average Fed speech requires a college degree (Siklos et al., 2018). As former Fed president Alan Greenspan quipped, "If you have understood me, I must have misspoken." The second problem disregards the primary purpose of language, the transmission of information from one person to another. Empirical studies of central bank communication have been remarkably cavalier about defining the public a central bank is supposed to address. This is all the more surprising because many of the most widespread macroeconomic models distinguish between different types of agents in the economy. It thus appears appropriate to proceed similarly when classifying the public central banks are communicating with.

Sentiment analysis has become a popular method to evaluate the effects of information contents on market actors. The dominant approach in the literature has been to employ dictionary methods, in which a set of n-grams are predefined and a document's sentiment score is calculated based on the relative occurrence of positive (e.g. "dovish") and negative (e.g. "hawkish") n-grams. This allows for interpretability and consistency. Picault and Renault (2017) (based on ECB press conferences), Bennani and Neuenkirch (2017) (ECB Governing Council speeches), create a field-specific hawkish/dovish sentiment dictionary to subsequently create time-series sentiment data on ECB speeches; other examples of dictionary techniques include Schmeling and Wagner (2015), Correa et al. (2017), Goldfarb et al. (2005), Stekler and Symington (2016) and Bennani (2020), the latter using an optimistic/pessimistic dictionary variant. Moniz and Jong (2014) use Naive Bayes, a supervised machine learning method, to group MPC minute sentences from the Bank of England by topic and then infer sentiment via a dictionary method. Popular sentiment dictionaries that use the hawkish/dovish distinction include Loughran and Mcdonald (2011) and Apel and Blix Grimaldi (2012). Some studies such as Gáti and Handlan (2022) focus on longer n-grams to capture more context. In all cases, there are no scores to capture the accuracy of n-gram classification methods other than manual verification of results. It is thus difficult to estimate if the produced sentiment scores truly reflect what the authors intended to constitute a positive/negative communication event.

A challenge of using bag-of-word techniques is to account for the linguistic idiosyncrasies of central bank communications. These have a notorious character, which central bankers such as Alan Greenspan used to wear like a crown and which have been laid aside only

[1] The SNA includes non-profit institutions (NPIs). Since these are usually not relevant for monetary policy, we have excluded this sector and added the central bank as a self-referential meta-sector.

recently (Blinder et al., 2022; Ferrara and Angino, 2022; Issing, 2019). Central bank communications highly depend on context, yet n-grams cannot capture this: the bigram "raise rates" flags both the sentences "we will not raise rates" and "we will raise rates", which are opposite in meaning. This is further aggravated as dictionaries typically exclude qualitative adjectives (e.g., Loughran and Mcdonald, 2011), thus leaving the inference of sentiment to the interpretation of nouns. Similarly, topic models like LDA may group communication about both negative and positive inflation shocks under the same topic, as these typically include the same terms (e.g. "CPI", "interest rates", "prices"). More sophisticated methods (e.g. using longer n-grams) to address such challenges have the adverse effect of substantially reducing data points, leading to less robust results when analyzing infrequent communications.

Another strand of literature has focused on topic models for analyzing central bank communications. Boukus and Rosenberg (2006) use Latent Semantic Analysis (LSA) to group FOMC statements by themes. Meade and Acosta (2015) and Ehrmann and Talmi (2020) use unweighted and TF-IDF-weighted document-term matrices to determine cosine-similarity between speeches. Following this topic model approach, Hansen and McMahon (2016) and Feldkircher et al. (2023) pursue an LDA-based (Blei et al., 2003) approach to group speeches by topics or ideological position, respectively. One weakness of topic-model approaches is that they do not directly provide an interpretable sentiment, but merely show connections within document term matrices. Further, data-driven methods do not allow for comparability between data sets (for example, clustering provided by a topic model only makes sense within the context of the data set), which makes it difficult to generalize empirical findings with regards to the contents of central bank communication.

A solution may be found in recent finance studies that show that machine learning algorithms outperform bag-of-word methods in sentiment analysis, including domain-specific dictionaries (Frankel et al., 2022; Purda and Skillicorn, 2015). Further, Huang et al. (2022) demonstrate that deep-learning models outperform non-deep-learning models in finance sentiment analysis. Such deep learning models, also referred to as large language models (LLMs), are typically pretrained on a large text corpus (Devlin et al., 2019; Peters et al., 2018; Radford et al., 2018) and then fine-tuned on a task-specific data set to extrapolate sentiment for out-of-sample data. Unlike dictionary approaches, LLMs detect more general patterns and can thus determine sentiment without a predefined n-gram present. LLMs have the additional advantage that they consider the context of a word in a given sentence, e.g., negation is detected. Thus far, LLMs have found application to the central bank communications literature only through their capacity to generate word embeddings, which are then used for downstream econometric tasks (Cai et al., 2021).

On the conceptual side, we observe that a large theoretical literature has underlined the heterogeneity of audiences in central bank communication (Blinder et al., 2022; Coibion et al., 2020; Vayid, 2013).[2] For example, Binder (2017) and Coibion and Gorodnichenko (2015) consider inflation expectations to be formed differently in households, firms and financial markets, which according to them would motivate more audience-specific communications from central banks. However, in their use of data, they only differentiate economic agents on the level of the listener not on the speaker, e.g. by conducting households surveys but keeping monetary policy announcements unspecified (Coibion et al., 2022). Reid and Siklos (2020) bemoan the lack of audience-discriminating data in central bank communications. While awareness for audience-specific analysis has thus received substantial theoretical support, it is yet to materialize in empirical analyses. Blinder et al. (2008) observe that "virtually all the research to date has focused on central bank communication with the financial markets" (p. 941). Meanwhile, Pfeifer et al. (2022) conduct an audience-based sentiment analysis and demonstrate that Fed communications distinguish between economic agents. This means that the *cui bono* question has largely been left unanswered by the literature: how valid is the classification of an event as positive or negative if heterogeneous audiences have opposing interests? The sentence "we will lower interest rates" may constitute a positive event for households who enjoy lower borrowing costs and a negative event for banks whose lending profitability is reduced.

If, then, one is to build on these considerations of audience-specific analysis, conceptual differences arise in the classification of central banks' target groups. Whom are central banks addressing? In Otmar Issing's *The Long Journey of Central Bank Communications*, one of first monographs dedicated to the subject, the author classifies ECB communications by target groups (Issing 2019, pp. 72–77). Among his target groups, one can find the general public, the media, markets, analysts, children, teenagers, and more. While this taxonomy provides insight into the audiences present at the announcement of a given communication, it does not say much about its mode of address. We find it more intuitive to classify target groups in central bank communications from the point of view of economic agents, i.e. those who may be directly affected by monetary policy. It is not possible to study the economic impact of monetary policy announcements without prior knowledge about what constitutes "the economy". The most basic taxonomy for classifying the economy into macroeconomic sectors is provided by the System of National Accounts (SNA 2008).[3] We will follow this taxonomy below and distinguish between households, firms, the financial sector, the government and the monetary authority. While it would certainly be insightful to gain knowledge from more granular classifications, e.g. by sub-classifying non-financial firms into "manufacturing", "agriculture", etc., more research is necessary to confirm the impact of central communications for these basic sectors.

Two problems exist within current NLP models used analyze central bank communications: the inability to consider contextual information–*context sensitivity*–and the lack of attention to communication as an interaction between the speaker and the audience–*audience specificity*. In the next section, we will construct two deep learning-based classification models with the hope that they will remedy these two shortcomings of measuring central communications.

[2] This field is in part driven by a new paradigm in central banking where targeted communications have gained importance. Over the last decade, central banks have thus increased their engagement on social media and implemented different strategies to convey information at varying levels of complexity, such as expanding the availability of educational materials on platforms such as websites or "listening events" (Blinder et al., 2022).

[3] Here, too, there are differences. For example, there is no link between the European Classification of Economic Activities (NACE), the Classification of Institutional Units in the System of National Accounts (SNA) or in the European System of Accounts (ESA). However, these differences concern the sectorial breakdown within the major sectors. Thus ESA (2013) uses the same five mutually exclusive sectors as SNA.

3. Training data and labeling methodology

Our corpus comprises the three largest English-speaking databases of central bank speeches, the BIS (1998–2022), which includes speeches of 83 central banks from Albania to Zambia; the Fed (1948–2022), including speeches of the board of governors and of the 12 district banks; and of the ECB (1999–2022), including speeches of the board members and of the 20 national member banks. The BIS data set consists of 10,211 speeches, the Fed data set of 6765 speeches and the ECB of 2405 speeches.[4] Unlike many NLP algorithms, deep-learning algorithms such as BERT do not require preprocessed inputs (e.g., removing stop words and punctuation, stemming, and lemmatizing). Instead, entire raw sentences are taken as inputs. We have thus kept preprocessing at a minimum by only separating each speech into individual sentences that have at least 20 characters.

In order to train our economic agent classifier, we manually labeled 6205 randomized sentences from the Fed database as speaking either about households, firms, the financial sector, the government, or the central bank itself. The question of accurately labeling sentences according to these relatively simple criteria can illustrate some of the ambiguity issues raised above. For instance, when a central banker speaks about mortgages, the information may be relevant to several actors at once. It could be relevant to households looking for home loans, to businesses looking for commercial loans, to the financial sector administering such loans or to the public sector, in the case of the government-backed mortgages. Lastly, in a context of supervisory oversight, it could pertain to the activities of the central bank itself. In each case, context is key to determining the semantic content of the sentence. We have therefore only labeled sentences where it is unambiguously clear that one and only one of our economic agents is being spoken about. Sentences about more than one economic agent have not received multiple labels.

The training data set for our sentiment classifier consists of 6683 manually prelabeled sentences from the Fed database, which are either labeled as being positive (1) or negative (0). We understand positive and negative sentiment as any sentence that talks positively or negatively about one and only one of our economic agents. We have thus only labeled the sentiment for sentences that had previously been attributed a label of one of the economic agents. Sentences that were flagged to speak about the central bank itself were excluded because they significantly reduced the accuracy of the model. One reason for this may be that sentences about the central bank are unfitting for binary classification; most often they include technical or operational statements. Positive and negative sentiment labels include descriptive sentences, e.g. sentences that simply portray the current, past or future economic situation of an economic agent, as well as prescriptive labels, e.g. sentences that make claims about how the economic situation should or should not be. Samples for both classification tasks are reported in Table 1.

We have generally found more negative labels during our manual labelling process and have kept these proportions in the data set. It may seem that this finding contradicts earlier studies which reported overall more positive communications (Apel and Blix Grimaldi 2012; Bennani and Neuenkirch 2017), even though we cannot infer from our labeling process that central bank communications are overall more negative. What we can say is that communications related to specific economic agents are, when taken together, more negative. Since we have not labeled sentences like "the economy is going well" for their lack of specificity, overall sentiment may be more than the sum of the parts. This discovery is noteworthy in itself, indicating that central banks might issue predominantly positive communications overall, while their targeted communications are more frequently negative. The reliance on bag-of-words in natural language processing methods of analyzing central bank communications may not have been able to accurately capture this apparent contradiction.

It would be tempting to suggest that our classifiers are not only able to assess when a central bank is talking *about* the economic situation and prospect of a given economic agent, but also *to* the agent. While we would be cautious in interpreting our economic agent classifier in this way, the sentiment classifier may be considered to fulfill this task. Because we labeled sentiments as positive or negative only with respect to an agent, the sentiment classifier is target-oriented, so that it is reasonable but not strictly necessary to assume that expressions of opinions and attitudes about an economic agent are also intended to be directed towards that agent. Forward guidance, the idea that communicating about the future course of monetary policy, can influence economic conditions today, is based on a similar assumption. It considers that whatever central banks are speaking about is a mode of address.

Table 1
Samples from manually prelabeled data set.

Speaker	Sentence	Sentiment	Economic Agent
M. H. Moskow	On a very practical level, we reestablished our competitiveness in financial services.	1	Financial Sector
R. P. Forrestal	Continued success on the economic front depends largely on political reform, namely constitutional reforms that will help restrain fiscal spending.	0	Government
R. W. Fisher	Our homeowners are thus less burdened with their housing predicament and better positioned as consumers.	1	Households
P. A. Volcker	Business investment has surged higher and productivity trends have shown sane signs of improvement from the poor record of the 1970s.	1	Firms
A. M. Santomero	In fact, payments processing is the largest component of fed operations.	Na	Central Bank

Note: Five samples from a manually prelabeled data set of FED speeches (1948–2022). Each row represents a labeled sentence along with speaker information, the sentence content, assigned sentiment label, and the corresponding economic agent label.

[4] Our labeled data sets can be found on GitHub: https://github.com/Moritz-Pfeifer/CentralBankRoBERTa/tree/main/Data/Labelled_data and on Huggingface.

We have created two additional data sets of pseudo-labels from the BIS and ECB data sets for each classifier. Pseudo-labeling is the process of adding confident predicted test data to training data (Riloff, 1996; Lee, 2013). To further fine-tune our classifiers, only pseudo-labels for with a high confidence threshold were used. For the economic agents classifier, a threshold of 77% for the BIS and 79% for the ECB was applied. As a result, 4804 additional labels were created for the BIS and 4051 for the ECB. This leaves us with a total of 15,060 labels for classifying economic agents in central bank communications. The training data set for the sentiment classifier contains an additional 2563 pseudo-labels generated from the ECB data set and an additional 4212 pseudo-labels generated from the BIS data set, both with a threshold of 87%. All together, we have 13,458 labels for our sentiment classifier, of which 7730 are negative and 5728 are positive. The proportion of each label reflects the proportion of the randomized sentences of the Fed data set. An overview of the training data for both classifiers is given in Fig. 1.

4. Experiments and results

In order to test which NLP algorithm is most appropriate to classify central bank communications, we fine-tune a variety of deep-learning models and other machine learning algorithms with our prelabeled training data. The presence of limited text data in fields such as central bank communications poses distinct challenges for building effective natural language processing systems. Creating high-quality datasets and testing for the appropriate model for a given domain-specific classification task mitigate some of these constraints. While costly in terms of human resources–accurate labeling requires specialized knowledge and is time consuming–it still provides the benchmark for most machine and deep learning tasks. For example, the standard movie review data set used for the General Language Understanding Evaluation (GLUE) benchmark of sentiment analysis is humanly annotated (Socher et al., 2013). Here, we aim to develop a robust and effective natural language processing system tailored to solve the challenges posed by our classification tasks: detecting the target groups and sentiment of central bank communications.

For the economic agent classifier, we fine-tune the prominent RoBERTa, BERT and XLNet (Yang et al., 2019) large language models. For a primer on the BERT-architecture that the RoBERTa (Devlin et al., 2019) and BERT large language models use, see Appendix A.1. We split the economic agent data set of 15,060 sentences into 80% (12,048) training and 20% (3012) validation data. The LLMs are trained to assign each sentence from our data set to one of five economic agents: households, firms, the financial sector, government and the central bank. The fine-tuning methodology is elaborated in Appendix A.2. We find that the RoBERTa model scores overall higher than both the BERT and XLNet models for our training task (see Table 3). A breakdown of the accuracy for each individual label shows that all three models achieve an accuracy of 90% or higher for all groups (see Table 7 in the Appendix). RoBERTa outperforms the other models only for sentences labeled as financial sector and government and achieves equal accuracy for other labels.

This outcome is particularly surprising given the inherent challenge of distinguishing between economic agents in sentences that often share similarities, such as the application of profits and losses to both the financial and non-financial corporate sectors or sentences discussing taxation policies that are applicable to both households and the government. More traditional bag-of-words approaches would struggle to discern economic agents solely based on terms like “taxes” or “profits”. The complexity of central bank communications, where similar terms are often used across different agents, poses a significant challenge that LLMs, with their capacity to capture contextual relationships and dependencies, thus successfully overcome. The high accuracy scores emphasize the effectiveness of LLMs in handling the problem of agent ambiguity displayed by central bank communications.

For the sentiment classifier, we fine-tune the RoBERTa, BERT, XLNet, and FinBERT large language models. We also consider other machine and deep learning approaches: Long Short-Term Memory (LSTM), Support Vector Machine (SVM), Random Forest, and Naïve Bayes (NB) classifier. LSTM is a basic type of recurrent neural network that can capture long-term dependencies (Hochreiter and Schmidhuber, 1997). SVM separates data into categories by constructing hyperplanes (Mullen and Collier, 2004). Random Forest Algorithms combine decision trees for data discrimination (Frankel et al., 2022). NB assigns a most likely label to a sentence using probabilistic reasoning based on Bayes' theorem (Li, 2010). These additional machine learning models were included in the baseline because they represent popular methods in the sentiment analysis literature and thus offer an additional point of comparison of our model with more traditional machine learning methods. The models are trained to assign each sentence from our data set a binary

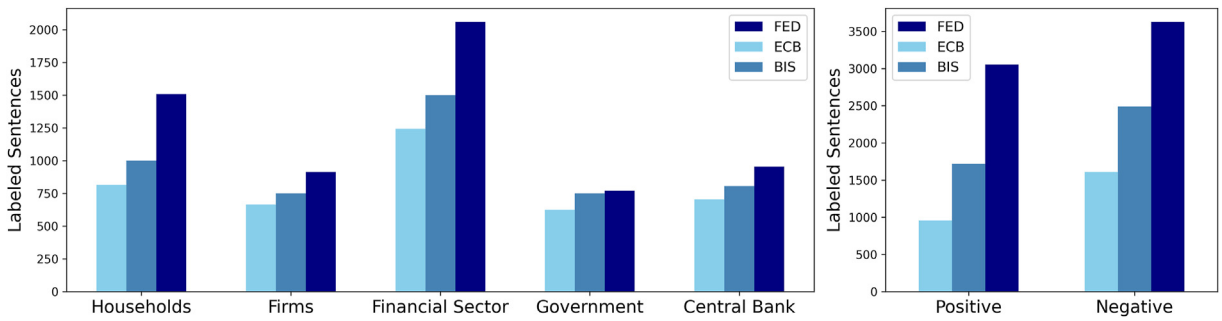

Fig. 1. Training Data.
Note: Overview of prelabeled training data. The economic agents classification data set (left) contains a total of 15,060 labels; the sentiment classification data set (right) 13,458 labels.

Table 2
Hyperparameters agent classifier.

Model	GAS	Batches	LR	Epochs
BERT	2	32	4e-6	5
XLNet	2	32	4e-6	5
RoBERTa	2	64	7e-6	5

Note: Hyperparameters used for the classification of different deep-learning algorithms applied to 3012 testing sentences labeled by the researchers. These hyperparameters include GAS (gradient accumulation), Batch size, LR (learning rate) and training epochs.

Table 3
Result agent classifier.

Model	Precision	Recall	F1
BERT	0.92	0.92	0.92
XLNet	**0.93**	0.92	0.92
RoBERTa	**0.93**	**0.93**	**0.93**

Note: Weighted averages of each model's Precision, Recall, and F1 score. Best results are bold. A detailed report on the performance of each label can be found in Table 7 of Appendix C.

Table 4
Hyperparameters sentiment classifier.

Model	GAS	Batches	LR	Epochs
LSTM	0	64	4e-4	3
BERT	2	64	2e-6	5
FinBERT	2	32	1e-6	5
XLNet	2	32	9e-7	5
RoBERTa	2	64	7e-6	5

Note: Hyperparameters used for the classification of different deep-learning algorithms applied to 2692 testing sentences labeled by the researchers. These hyperparameters include GAS (gradient accumulation), Batch size, LR (learning rate) and training epochs.

Table 5
Result sentiment classifier.

Model	Precision	Recall	F1
NB	0.82	0.76	0.74
RF	0.80	0.80	0.79
LSTM	0.81	0.81	0.81
SVM	0.83	0.83	0.83
FinBERT	0.85	0.85	0.85
BERT	0.85	0.85	0.85
XLNet	0.87	0.87	0.87
RoBERTa	**0.88**	**0.88**	**0.88**

Note: Weighted averages of different machine-learning and deep-learning models, Precision, Recall, and F1 score. Best results are bold. A detailed report on the performance of each label can be found in Table 8 of Appendix C.

positive/negative label. We split the sentiment data set of 13,458 sentences into 80% (10,766) training and 20% (2692) validation data. We elaborate on the fine-tuning methodology for LLMs in Appendix A.2 and in Appendix B for the other models.

Our findings (see Table 5) show that all LLMs outperform all non-LLM models for our task. Among the LLMs, the RoBERTa model outperforms the FinBERT, BERT, and XLNet fine-tuned models. The RoBERTa model also outperforms all other models on the level of individual accuracy for each label (see Table 8). Breaking down the accuracy for each label, we note that all models display a slightly higher accuracy for negative labels compared to positive labels, ranging from a 21% difference for Naïve Bayes to a 3% difference for RoBERTa. This may be due to the larger sample size of negative labels or to the inherent challenge in accurately capturing positive instances, which may be more nuanced and context-dependent.

As the RoBERTa model strictly outperforms all other models in both classification tasks, we release the fine-tuned economic agent classifier and sentiment classifier based on the RoBERTa model to the public under the name CentralBankRoBERTa.[5] The architecture of CentralBankRoBERTa is elaborated in Appendix A.2.

It should be noted that CentralBankRoBERTa does not include domain-specific pretraining. This is in contrast to the FinBERT model (Huang et al., 2022), a similarly domain-specific application of the BERT architecture for financial texts. Pretraining a BERT-based algorithm requires a large data set of domain-specific text so that the algorithm can infer the meaning of words as they are used in

[5] We release our model, data and code on GitHub: https://github.com/Moritz-Pfeifer/CentralBankRoBERTa.

that domain. Pretraining is of particular use when labeled training data for fine-tuning is sparse (Huang et al., 2022). We forgo this pretraining step because our labeled data set for fine-tuning is sufficiently large to justify the usage of the general pretraining provided by the RoBERTa model. Further, the fine-tuned FinBERT exhibits no performance increase (see Tables 5 and 8appsec3) over the BERT model for sentiment classification, which is noteworthy, given that the field of central banking is considered adjacent to finance. Lastly, pretraining requires more data than is currently available in the domain of central bank communication. Taken together, the unlabeled BIS, FED and ECB speeches only represent 477 MB. Nevertheless, future research might further improve CentralBankRoBERTa's result by adding a pretraining step that could arguably further improve the accuracy of the model. Data potentially useful for pretraining could include press releases, minutes, congressional hearings, policy briefings as well as monetary policy news.

5. Conclusion

The literature analyzing central bank communications has relied almost exclusively on bag-of-words methods that ignore contextual information. These methods have shortcomings that limit their ability to capture the nuanced meaning and subtleties inherent in central bank communications. In spite of efforts to communicate more clearly, central bank communications remain intricate and not easily decipherable by standard NLP methods. This calls for context-sensitive text analysis methods. Additionally, a nascent literature focuses on differentiating target groups in the analysis of central bank communications. This calls for text analysis methods that can discriminate audiences on the speaker level. We introduce CentralBankRoBERTa, a state-of-the-art LLM based on the pretrained RoBERTa model. We fine-tune the model for two tasks: (1) classification of economic agents and (2) sentiment in central bank communication. We train our model on two unique sets of manually prelabeled sentences from Fed, ECB and BIS communications, marking a novel effort in creating a data set tailored specifically for economic agent classification in central bank communications. The manual annotation of the data set enables our model to discern the complexities inherent in central bank communications that may be overlooked by traditional bag-of-words methods. We benchmark CentralBankRoBERTa on other LLMs (BERT, FinBERT, XLNet) and machine learning techniques (NB, SVM, RF) and find it outperforms them in both tasks.

CentralBankRoBERTa provides a novel way to account for context in central bank communication. Our economic agent classifier may be useful for research interested in understanding the ways in which central bank communications address different actors within the economy. This can refine previous analyses by discriminating the effects of communication on market events by separating the addressed economic agent. It also allows researchers and policymakers to gain insights into whether certain actors are being favored or disadvantaged by the central bank's communications and, by extension, monetary policy. By opening up new ways to identify the central bank's audiences, researchers and policymakers can seek to answer questions such as whether communication to ordinary citizens is effective or a failed endeavor. In this context, CentralBankRoBERTa may be used as a tool to assess the progress that central banks have made regarding the goal of reaching a “broader audience” (e.g., ECB, 2020).

The sentiment classifier can be a valuable tool for researchers interested in monitoring how central banks feel about the various constituents of the economy, and whether these feelings influence economic outcomes. While the empirical literature for the economic impact of central bank communications on financial markets has grown into a burgeoning sub-discipline, evidence for their economic effect on households, firms and fiscal policy is still an evolving area of study. Since CentralBankRoBERTa classifies sentiment with respect to economic conditions (as opposed to Hawkish/Dovish policy stances), it can also be applied to other contexts where economic agent classification is pertinent. Fruitful applications may include the quantification of firms' outlook on economic conditions or the fiscal authority's sentiment about different constituents of the economy. Further, CentralBankRoBERTa's two tasks synergize, and researchers may for the first time analyze group-specific sentiment, which can spawn novel economic analyses.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. CentralBankRoBERTa's architecture

A.1. A primer on BERT-based models

BERT, introduced in 2019 by Devlin et al., stands for Bidirectional Encoder Representations from Transformers. It differs from prior NLP methods by considering the context of a word in both directions (left and right) simultaneously, which allows BERT to better capture the nuances of natural language. BERT processes tokenized inputs (for the $\text{BERT}_{\text{Base}}$ model) using a 12-layer bidirectional Transformer encoder (Vaswani et al., 2017), outputting a list of 768-dimensional vectors (one vector per input token), which can then be used for downstream tasks. Since publication, numerous models building on the original BERT architecture have been released.

CentralBankRoBERTa is a fine-tuned version of the RoBERTa model. RoBERTa (Liu et al., 2019) is a more streamlined version of BERT that outperforms it in common benchmarks (GLUE, SQuAD). Notably, it removes BERT's Next Sentence Prediction (NSP) feature and operates with larger mini-batch sizes, learning rates, and a larger pretraining data set of 160 GB compared to BERT's 16 GB (Liu et al., 2019).

The other LLMs used in our experiment are the FinBERT and XLNet model. FinBERT (Huang et al., 2022) is a BERT-based model whose distinguishing feature is its inclusion of a large corpus of financial texts (such as 10-K/10-Q forms, earning call transcripts and analyst reports) in its pretraining. The advantage of such pretraining is that FinBERT can achieve higher accuracy scores for

finance-specific domain applications with less fine-tuning data. However, differences become marginal as the fine-tuning data set becomes large (Huang et al., 2022). The XLNet large-language model is not based on the BERT architecture. XLNet operates as an autoregressive language model, generating the joint probability of a sequence of tokens using the transformer architecture with recurrence. In its training process, the model computes the probability of a word token, taking into account all permutations of word tokens within a sentence rather than solely those positioned to the left or right of the target token (Yang et al., 2019). XLNet outperforms BERT and RoBERTa in some benchmarks such as GLUE and SQuAD, and was therefore included in our experiment.

A.2. Fine-Tuning CentralBankRoBERTa

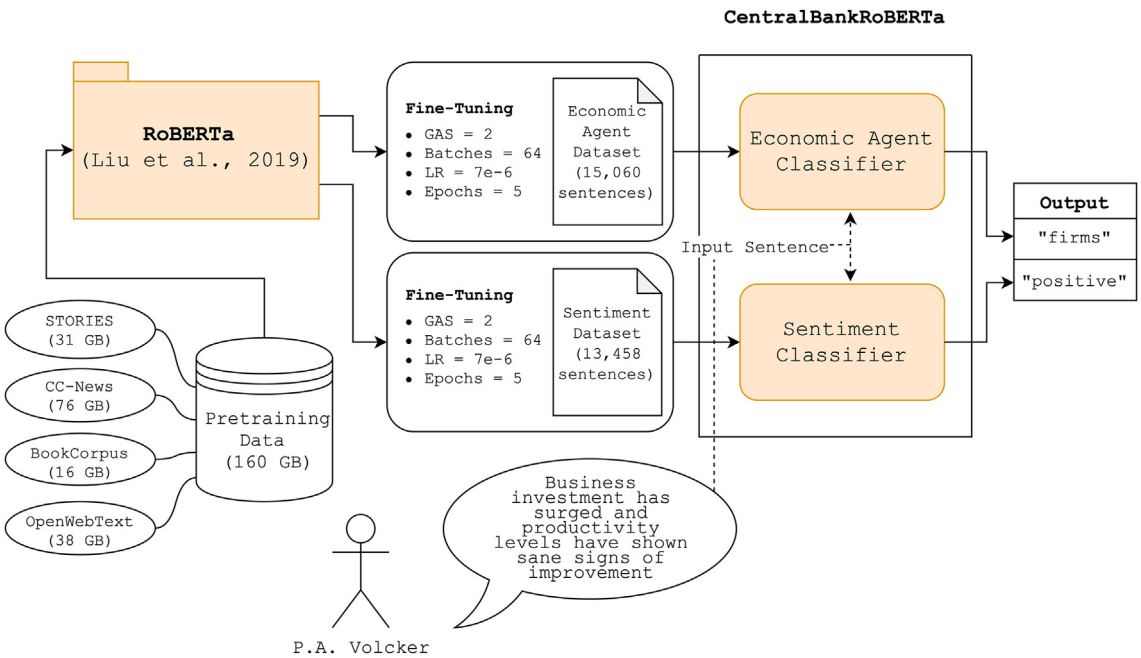

Fig. 2. CentralBankRoBERTa Architecture.
Note: A visualization of the CentralBankRoBERTa training pipeline. We use the pretrained RoBERTa model and fine-tune it using our data sets on Economic Agents and Sentiment, respectively.

Training a BERT-based model requires two stages: pretraining and fine-tuning. During pretraining, the model can learn a broad understanding of language structure and context by being exposed to a large corpus of text. This foundation is crucial for the model's ability to grasp nuances and perform well in specific tasks during the fine-tuning stage. As argued above, we forgo pretraining a BERT-based model from scratch, and instead use the pretraining provided by the respective BERT, RoBERTa, and FinBERT models. We also keep the pretraining of the XLNet model. We visualize the training pipeline for the final CentralBankRoBERTa model based on the RoBERTa architecture in Fig. 2. However, the training pipeline can be understood homologously for the BERT, FinBERT and XLNet models.

Table 6
Overview of Pretraining Data for LLMs

Model Name	Training Data Type	Training Data Size
BERT	BookCorpus, Wikipedia	16 GB
RoBERTa	BookCorpus, CC-News, OpenWebText, STORIES	160 GB
FinBERT	10-K and 10-Q forms, Thomson Investext Analyst Reports, Earnings Conference Call transcripts	~20–30 GB
XLNet	BookCorpus, Wikipedia, Giga5, ClueWeb, Common Crawl	158 GB

Note: *Training Data Type* refers to the data used in the model's pretraining step. The column *Training Data Size* indicates the size of the uncompressed text size in gigabytes. The data set size for FinBERT is estimated as Huang et al. (2022) only provide the number of tokens used in their paper.

For our experiment, we use the PyTorch implementation of the RoBERTa, BERT, FinBERT, and XLNet models via the HuggingFace transformer open-source library. We use the base versions of all models because performance increases of the large versions were observed to be marginal and could therefore not justify the additional computing costs. All training and testing was carried out on a A100 SXM4 GPU with 40 GB memory within the Google Colab environment.

To increase the batch size and still be able to fit higher batch sizes into memory, we add gradient accumulation steps (GAS) to our model. For all models and both classification tasks, warmup steps (n = 500), gradient accumulation steps (n = 2) and decay weight

(0.01) parameters are set equal. Dropout rates and parameter weights are left at the default settings for all models. We fine-tune the large language models by testing parameters for batch size, learning rate (LR) and epochs. Manual testing has yielded the hyperparameters in Table 2 to be optimal for the Economic Agent classifier and the hyperparameters in Table 4 to be optimal for the Sentiment classifier.

Appendix B. Implementation of NB, RF, LSTM and SVM Algorithms

Given that there exists an established literature on NLP for sentiment classification, we include common deep and machine learning techniques in our experiment: Long Short-Term Memory (LSTM), Support Vector Machine (SVM), Random Forest, and a Naïve Bayes (NB) classifier.

For the LSTM model, we use the implementation provided by the Python package keras. We determine optimal parameters for the model via the Ax platform (ax.dev). The LSTM uses tokenized inputs which we generate with the tokenizer implementation of keras. For more detail on the LSTM model, see Hochreiter and Schmidhuber (1997).

The inputs for the SVM, Random Forest and NB classifiers are the vector outputs of the TF-IDF algorithm. TF-IDF, short for term frequency–inverse document frequency, generates a vector that represents how important a word is in a collection or corpus. We use the TF-IDF implementation from the scikit-learn Python package.

We use the scikit-learn implementations of the SVM, Random Forest and NB classifiers for our experiment. We fine-tune the parameters of the SVM model using the gridsearchCV function from the scikit-learn package. We keep default parameters for the Random Forest provided by the scikit-learn package. NB only has one parameter α, which we determine by cycling through the possible input range between 0.01 and 100 in increments of 0.01. We determine $\alpha = 0.28$ to be the optimum value. All details on the training can be found on Github.

Appendix C. Additional Tables

Table 7
Detailed Result Agent Classifier

Model	Precision	Recall	F1	Support
BERT				
Households	0.95	0.91	0.93	665
Firms	0.90	0.90	0.90	465
Financial Sector	0.93	0.95	0.94	961
Government	0.92	0.92	0.92	429
Central Bank	0.93	0.93	0.93	492
XLNet				
Households	0.95	0.91	0.93	665
Firms	0.90	0.89	0.90	465
Financial Sector	0.91	0.96	0.94	961
Government	0.95	0.91	0.93	429
Central Bank	0.93	0.93	0.93	492
RoBERTa				
Households	0.96	0.90	0.93	665
Firms	0.87	0.92	0.90	465
Banks	0.94	0.96	0.95	961
Government	0.95	0.93	0.94	429
Central Bank	0.92	0.95	0.93	492

Note: Classification performance of different deep-learning algorithms for testing sentences labeled by the researchers. The table reports each model's Precision, Recall, F1 score and Support for each label.

Table 8
Detailed Result Sentiment Classifier

Model	Precision	Recall	F1	Support
Naive Bayes				
Negative	0.71	0.99	0.83	1546
Positive	0.96	0.46	0.62	1146
Random Forest				
Negative	0.79	0.87	0.83	1546
Positive	0.80	0.69	0.74	1146
LSTM				
Negative	0.79	0.89	0.84	1546
Positive	0.84	0.70	0.76	1146
Support Vector Machine				
Negative	0.84	0.88	0.86	1546
Positive	0.82	0.77	0.79	1146
BERT				
Negative	0.86	0.90	0.88	1546
Positive	0.85	0.80	0.83	1146

(*continued on next page*)

Table 8 (*continued*)

Model	Precision	Recall	F1	Support
FinBERT				
Negative	0.85	0.90	0.88	1546
Positive	0.86	0.79	0.82	1146
XLNet				
Negative	0.88	0.89	0.89	1546
Positive	0.85	0.84	0.84	1146
RoBERTa				
Negative	0.90	0.90	0.90	1546
Positive	0.86	0.87	0.87	1146

Note: Classification performance of different machine-learning and deep-learning algorithms for testing sentences labeled by the researchers. The table reports each model's Precision, Recall, F1 score and Support for each label.

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